University of Moratuwa

Department of Electronic and Telecommunication Engineering

UOM Backbone and ENTC Local Area Network Design

Network Architecture, Routing Simulation, and BOQ

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1 Introduction

This report presents the design and simulation of the backbone network for the University of Moratuwa and the internal Local Area Network (LAN) of the Department of Electronic and Telecommunication Engineering (ENTC). The backbone network interconnects all major buildings on campus through a hierarchical topology centered on the Network Operations Center (NOC) at the Center for Information Technology Services (CITeS).

The network has been simulated using Cisco Packet Tracer, with OSPF configured for dynamic routing across all backbone nodes.

2 Backbone Network Design

2.1 Approach and Topology Selection

The backbone network follows a **hierarchical star topology with partial mesh redundancy**. The NOC (CITeS) acts as the core node, with distribution-level buildings (ENTC, Faculty of IT) acting as aggregation points for clusters of access-level buildings. This design was chosen for the following reasons.

- **Scalability:** New buildings can be added by connecting to the nearest distribution node without redesigning the core.
- **Cost efficiency:** Fiber runs are minimized by directing traffic through intermediate nodes rather than running direct links to the NOC from every building.
- **Redundancy:** A partial ring exists in the topology (NOC – ENTC – Admin – Mechanical – Material and ERE – Sumanadasa – Textile – Civil – FIT – NOC), providing alternate paths in the event of a single link failure.
- **Performance:** All links operate at 1 Gbps over single-mode fiber, sufficient for a 20–25 year design lifetime.

OSPF (Open Shortest Path First) is used as the routing protocol. OSPF is an industry-standard link-state protocol that automatically computes optimal paths, adapts to topology changes, and scales well to a network of this size.

2.2 Network Diagram

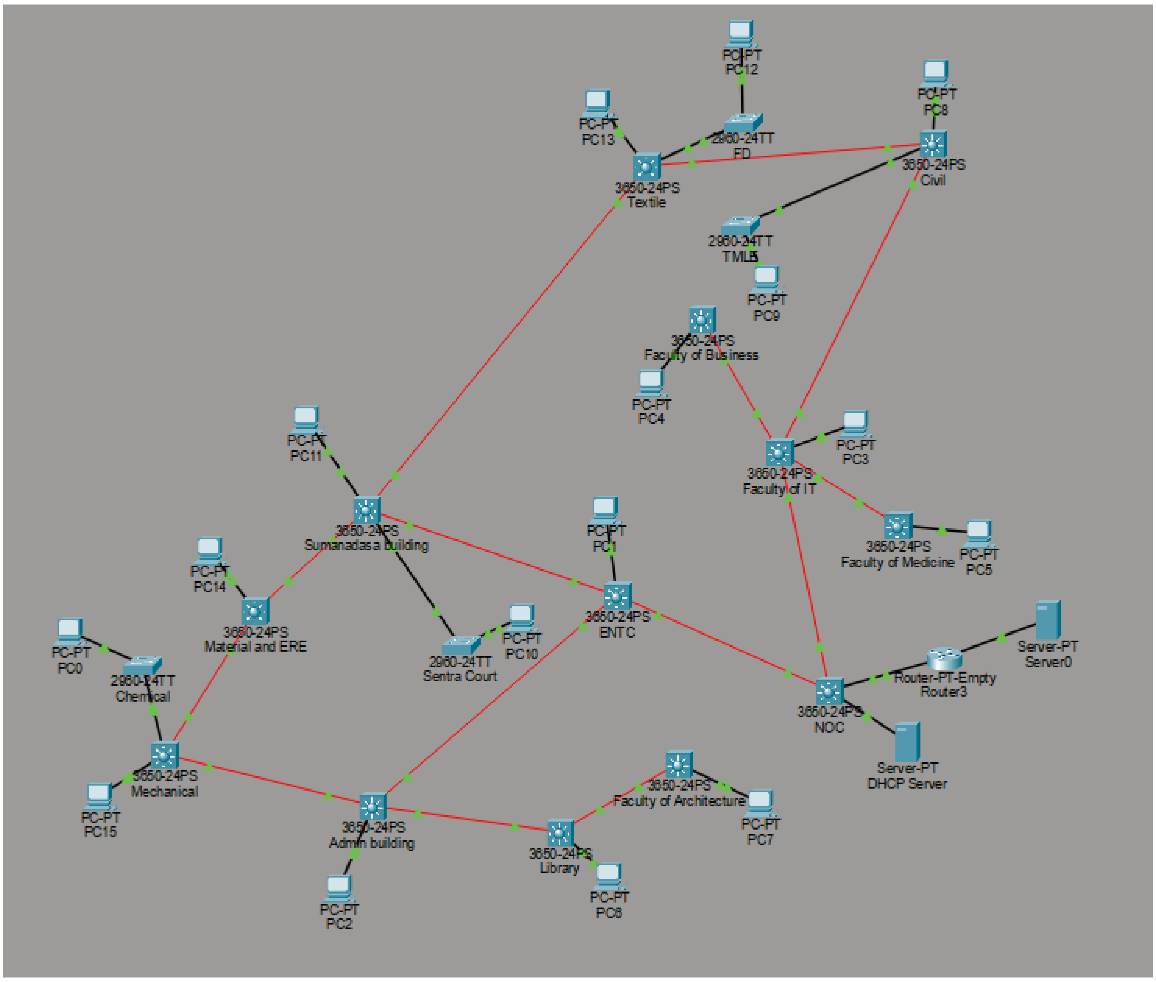

Figure 1: UOM Backbone Network Topology (Cisco Packet Tracer)

2.3 Building Nodes and Link Bandwidths

All inter-building links operate at **1 Gbps** over fiber optic and copper straight-through cables. Table 1 below lists all backbone links.

Table 1: Backbone Network Links

Node A	**Node B**	**Bandwidth**	**Medium**
NOC	ENTC	1 Gbps	Single-mode fibre
NOC	FIT	1 Gbps	Single-mode fibre
ENTC	Admin	1 Gbps	Single-mode fibre
ENTC	Sumanadasa	1 Gbps	Single-mode fibre
FIT	FOB	1 Gbps	Single-mode fibre
FIT	FOM	1 Gbps	Single-mode fibre
FIT	Civil	1 Gbps	Single-mode fibre
Admin	Library	1 Gbps	Single-mode fibre
Admin	Mechanical	1 Gbps	Single-mode fibre
Library	FOA	1 Gbps	Single-mode fibre

Node A	Node B	Bandwidth	Medium
Civil	Textile	1 Gbps	Single-mode fibre
Civil	TMLE	100 Mbps	Cat6 UTP copper
Textile	Sumanadasa	1 Gbps	Single-mode fibre
Textile	FD	100 Mbps	Cat6 UTP copper
Sumanadasa	MSE	1 Gbps	Single-mode fibre
Sumanadasa	Sentra Court	100 Mbps	Cat6 UTP copper
MSE	Mechanical	1 Gbps	Single-mode fibre
Mechanical	Chemical	100 Mbps	Cat6 UTP copper
NOC	Router	1 Gbps	Single-mode fibre

2.4 IP Addressing Scheme

2.4.1 Design Rationale

A **/22** subnet per building was chosen, providing 1022 usable host addresses per building. This accommodates current needs and leaves room for growth. Inter-building links use **/30** point-to-point subnets from the `10.255.0.0/24` block, minimizing address wastage on transit links.

2.4.2 IPv4 Addressing

Table 2: IPv4 Building Subnet Allocation

Building	Subnet	Gateway	DHCP Start
NOC (CITeS)	10.0.0.0/22	10.0.0.1	10.0.0.10
ENTC	10.0.4.0/22	10.0.4.1	10.0.4.10
FIT	10.0.8.0/22	10.0.8.1	10.0.8.10
Admin	10.0.12.0/22	10.0.12.1	10.0.12.10
FOB	10.0.16.0/22	10.0.16.1	10.0.16.10
FOM	10.0.20.0/22	10.0.20.1	10.0.20.10
Library	10.0.24.0/22	10.0.24.1	10.0.24.10
FOA	10.0.28.0/22	10.0.28.1	10.0.28.10
Civil	10.0.32.0/22	10.0.32.1	10.0.32.10
Sumanadasa	10.0.36.0/22	10.0.36.1	10.0.36.10
Textile	10.0.40.0/22	10.0.40.1	10.0.40.10
MSE	10.0.44.0/22	10.0.44.1	10.0.44.10
Mechanical	10.0.48.0/22	10.0.48.1	10.0.48.10
TMLE	10.0.32.0/22	10.0.32.1	10.0.32.10 (shared with Civil)
Sentra Court	10.0.36.0/22	10.0.36.1	10.0.36.10 (shared with Sumanadasa)
FD	10.0.40.0/22	10.0.40.1	10.0.40.10 (shared with Textile)
Chemical	10.0.48.0/22	10.0.48.1	10.0.48.10 (shared with Mechanical)
Main Server	192.168.1.0/24	192.168.1.1	Static

Table 3: Point-to-Point Link IP Allocation

Link	Subnet	Node A IP	Node B IP
NOC – ENTC	10.255.0.0/30	10.255.0.1	10.255.0.2
NOC – FIT	10.255.0.4/30	10.255.0.5	10.255.0.6
NOC – Router	172.16.0.0/30	172.16.0.1	172.16.0.2
ENTC – Admin	10.255.0.8/30	10.255.0.9	10.255.0.10
ENTC – Sumanadasa	10.255.0.32/30	10.255.0.33	10.255.0.34
FIT – FOB	10.255.0.12/30	10.255.0.13	10.255.0.14
FIT – FOM	10.255.0.16/30	10.255.0.17	10.255.0.18
FIT – Civil	10.255.0.28/30	10.255.0.29	10.255.0.30
Admin – Library	10.255.0.20/30	10.255.0.21	10.255.0.22
Admin – Mechanical	10.255.0.52/30	10.255.0.53	10.255.0.54
Library – FOA	10.255.0.24/30	10.255.0.25	10.255.0.26
Civil – Textile	10.255.0.36/30	10.255.0.37	10.255.0.38
Textile – Sumanadasa	10.255.0.40/30	10.255.0.41	10.255.0.42
Sumanadasa – MSE	10.255.0.44/30	10.255.0.45	10.255.0.46
MSE – Mechanical	10.255.0.48/30	10.255.0.49	10.255.0.50

2.4.3 IPv6 Addressing

For IPv6, a `/48` organisational prefix would be requested from APNIC (the Regional Internet Registry for the Asia-Pacific region, which includes Sri Lanka). From this `/48`, each building is allocated a `/64` subnet, providing effectively unlimited host addresses per building. Using `2001:db8:uom::/48` as a placeholder (to be replaced with the actual APNIC-assigned prefix):

Table 4: IPv6 Subnet Allocation

Building	IPv6 Subnet
NOC	2001:db8:uom:0001::/64
ENTC	2001:db8:uom:0002::/64
FIT	2001:db8:uom:0003::/64
Admin	2001:db8:uom:0004::/64
FOB	2001:db8:uom:0005::/64
FOM	2001:db8:uom:0006::/64
Library	2001:db8:uom:0007::/64
FOA	2001:db8:uom:0008::/64
Civil	2001:db8:uom:0009::/64
Sumanadasa	2001:db8:uom:000a::/64
Textile	2001:db8:uom:000b::/64
MSE	2001:db8:uom:000c::/64
Mechanical	2001:db8:uom:000d::/64
TMLE	2001:db8:uom:0009::/64 (shared with Civil)
Sentra Court	2001:db8:uom:000a::/64 (shared with Sumanadasa)
FD	2001:db8:uom:000b::/64 (shared with Textile)
Chemical	2001:db8:uom:000d::/64 (shared with Mechanical)

2.5 Active Components – Selection and Justification

2.5.1 Backbone Switches – Cisco Catalyst 3650-24PS

All backbone building nodes use the Cisco Catalyst 3650-24PS Layer 3 switch. This was selected because it supports full Layer 3 routing including OSPF, has SFP uplink ports for fiber connectivity, and provides 24 copper ports for end devices.

Each switch acts as both a distribution switch for its building LAN (via VLAN 1 SVI as default gateway) and a backbone routing node (via routed SFP ports carrying /30 point-to-point links).

Key specifications: 24x 10/100/1000 PoE+ ports, 4x 1G SFP uplinks, 88 Gbps switching capacity, supports RIP, OSPF, EIGRP, and static routing.

2.5.2 Core Router – Router-PT (Real-world equivalent: Cisco 4321)

A Router-PT device is used in the Packet Tracer simulation to represent the campus core router. In a real deployment, a Cisco 4321 Integrated Services Router would be used. It connects the campus backbone to the main server subnet (`192.168.1.0/24`) and injects a default route into OSPF so all campus hosts can reach external resources. The Cisco 4321 supports 2 WAN GE ports, 2 NIM slots for expansion, and hardware encryption for VPN, all of which are suitable for a university internet gateway.

2.5.3 Centralized DHCP Server

A single DHCP server at the NOC serves all building subnets. Each building switch uses `ip helper-address` to relay DHCP discover packets to the server. This centralizes IP address management and simplifies administration.

2.5.4 Secondary Switches – Cisco Catalyst 2960-24TT

Buildings with secondary access switches (Chemical, TMLE, Sentra Court, FD) use the Cisco Catalyst 2960-24TT, a cost-effective Layer 2 switch. These rely on the upstream 3650 for routing and DHCP relay.

2.6 Passive Components – Selection and Justification

2.6.1 Fiber Type

Single-mode fiber (OS2) is used for all inter-building backbone links. UOM campus spans distances of up to several hundred meters between buildings, and the backbone must support a 20–25 year lifespan. OS2 single-mode fiber supports 10 Gbps over 10 km, making it future-proof for bandwidth upgrades without replacing the physical plant. Multimode fiber (limited to 550 m at 1 Gbps on OM3) is not suitable for a long-term backbone investment.

2.6.2 Connectors and Patch Panels

LC connectors are used throughout as the industry standard for high-density fiber termination. 12-port fiber patch panels with LC adapters are installed at each building's distribution rack. Copper horizontal cabling uses Cat6 UTP terminated at RJ45 keystone jacks and 24-port copper patch panels.

3 Bill of Quantities (BOQ)

3.1 ENTC LAN – Active Components

Item	**Description / Specification**	**Est. Qty**	**Unit Price**	**Total Price**
Core Switch	Cisco Catalyst 3650-24PS (MDF / Ground Floor)	1	$3,500.00	$3,500.00
Access Switch	Cisco Catalyst 2960-X (IDFs / Floor Switches)	12	$1,200.00	$14,400.00
Wireless AP	Cisco Aironet / Catalyst Wireless Access Points	10	$450.00	$4,500.00
			Subtotal	**$22,400.00**

Table 5: Active Network Equipment (ENTC Department)

3.2 ENTC LAN – Passive Components

Item	**Description / Specification**	**Est. Qty**	**Unit Price**	**Total Price**
Vertical Cabling	Multimode Fiber Optic Cable (Internal Backbone)	300 m	$2.50	$750.00
Horizontal Cabling	Cat6a UTP Copper Cable (100m per floor limit)	15 boxes	$200.00	$3,000.00
Patch Panels	24-Port Ethernet Patch Panels (MDF & IDFs)	13 units	$100.00	$1,300.00
Network Racks	Wall-mount / Floor-standing Cabinets	5 units	$400.00	$2,000.00
Wall Sockets	Dual RJ45 Ethernet Wall Outlets	120 units	$15.00	$1,800.00
			Subtotal	**$8,850.00**

Table 6: Passive Infrastructure Materials (ENTC Department)

3.3 Backbone – Active Components

Item	Description / Specification	Est. Qty	Unit Price	Total Price
Core Switch	Cisco Catalyst 3650-24PS (Layer 3, PoE+)	13	$3,500.00	$45,500.00
Access Switch	Cisco Catalyst 2960-X 24-Port (Layer 2)	4	$1,200.00	$4,800.00
Edge Router	Cisco 4000 Series Integrated Services Router	1	$2,500.00	$2,500.00
Enterprise DHCP Server	Rackmount Server (Dell PowerEdge R450 / Cisco UCS C220)	1	$5,000.00	$5,000.00
Enterprise Edge Server	High-Performance Rackmount (e.g., Dell PowerEdge R750 / Cisco UCS C240)	1	$12,500.00	$12,500.00
			Subtotal	**$70,300.00**

Table 7: Active Network Equipment (Campus Backbone)

3.4 Backbone – Passive Components

Item	Description / Specification	Est. Qty	Unit Price	Total Price
Backbone Cabling	Single-Mode Fiber Optic Cable (OS2)	2 km	$1,500.00	$3,000.00
Internal Cabling	Cat6a UTP Copper Cable	10 boxes	$200.00	$2,000.00
Termination	24-Port Fiber Patch Panels & LC Connectors	15 units	$150.00	$2,250.00
			Subtotal	**$7,250.00**

Table 8: Passive Infrastructure Materials (Campus Backbone)

3.5 Simulation Results

The following screenshots demonstrate successful simulation of both networks.

Figure 2 shows the complete backbone topology in Packet Tracer.

Figure 3 shows the output of `show ip route ospf` on the NOC switch, confirming that all building subnets are learned via OSPF.

Figure 4 shows a successful transmission from a PC in the FD building to a PC in the Sumanadasa building demonstrating end-to-end connectivity across the backbone.

Figure 5 shows a PC's IP configuration after obtaining an address via DHCP, confirming the DHCP relay is working correctly across buildings.

Figure 6 shows the ENTC internal network topology.

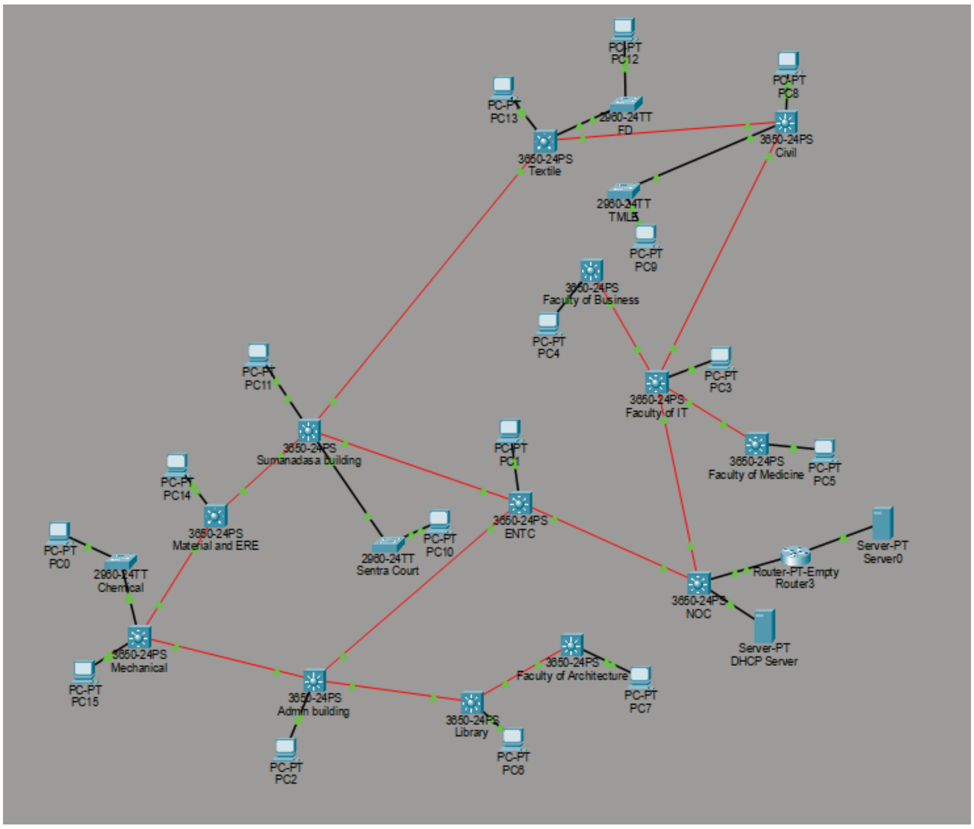

Figure 2: UOM Backbone Topology in Packet Tracer

```
NOC>show ip route ospf
     10.0.0.0/8 is variably subnetted, 30 subnets, 3 masks
O       10.0.32.0 [110/3] via 10.255.0.6, 02:23:10, GigabitEthernet1/1/2
O       10.0.36.0 [110/3] via 10.255.0.2, 02:12:03, GigabitEthernet1/1/1
O       10.0.40.0 [110/4] via 10.255.0.2, 02:04:11, GigabitEthernet1/1/1
                  [110/4] via 10.255.0.6, 02:04:11, GigabitEthernet1/1/2
O       10.0.44.0 [110/4] via 10.255.0.2, 01:59:35, GigabitEthernet1/1/1
O       10.0.48.0 [110/4] via 10.255.0.2, 01:59:20, GigabitEthernet1/1/1
O       10.255.0.8 [110/2] via 10.255.0.2, 02:41:22, GigabitEthernet1/1/1
O       10.255.0.12 [110/2] via 10.255.0.6, 02:41:22, GigabitEthernet1/1/2
O       10.255.0.16 [110/2] via 10.255.0.6, 02:41:22, GigabitEthernet1/1/2
O       10.255.0.20 [110/3] via 10.255.0.2, 02:41:04, GigabitEthernet1/1/1
O       10.255.0.24 [110/4] via 10.255.0.2, 02:41:04, GigabitEthernet1/1/1
O       10.255.0.28 [110/2] via 10.255.0.6, 02:23:20, GigabitEthernet1/1/2
O       10.255.0.32 [110/2] via 10.255.0.2, 02:12:13, GigabitEthernet1/1/1
O       10.255.0.36 [110/3] via 10.255.0.6, 02:04:21, GigabitEthernet1/1/2
O       10.255.0.40 [110/3] via 10.255.0.2, 02:04:11, GigabitEthernet1/1/1
O       10.255.0.44 [110/3] via 10.255.0.2, 01:59:35, GigabitEthernet1/1/1
O       10.255.0.48 [110/4] via 10.255.0.2, 01:59:20, GigabitEthernet1/1/1
O       10.255.0.52 [110/3] via 10.255.0.2, 01:59:20, GigabitEthernet1/1/1
O    192.168.1.0 [110/2] via 172.16.0.2, 02:41:35, GigabitEthernet1/0/2
```

Figure 3: OSPF route table on NOC – all building subnets learned via OSPF

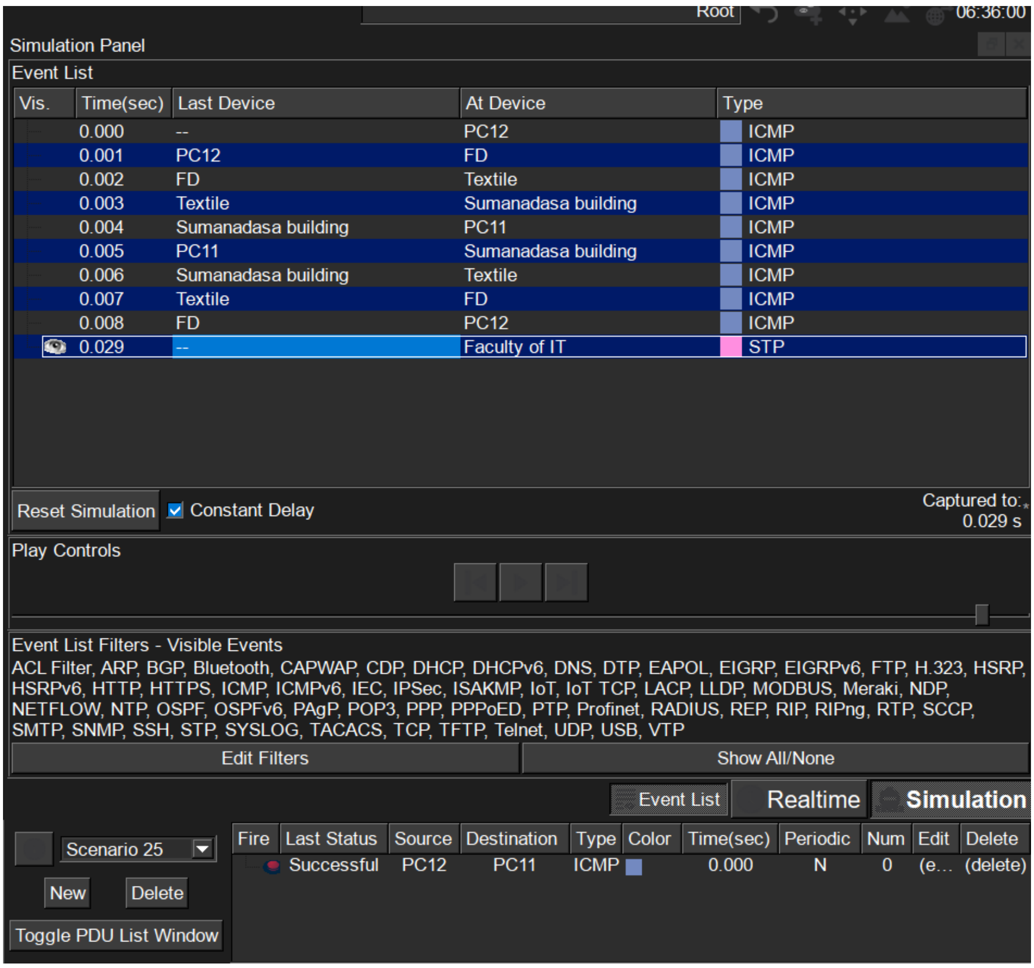

Figure 4: Successful PDU from FD PC to Sumanadasa PC

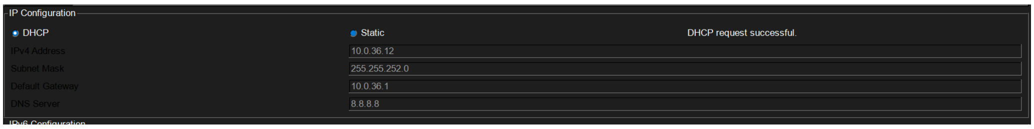

Figure 5: PC obtaining IP address via DHCP relay

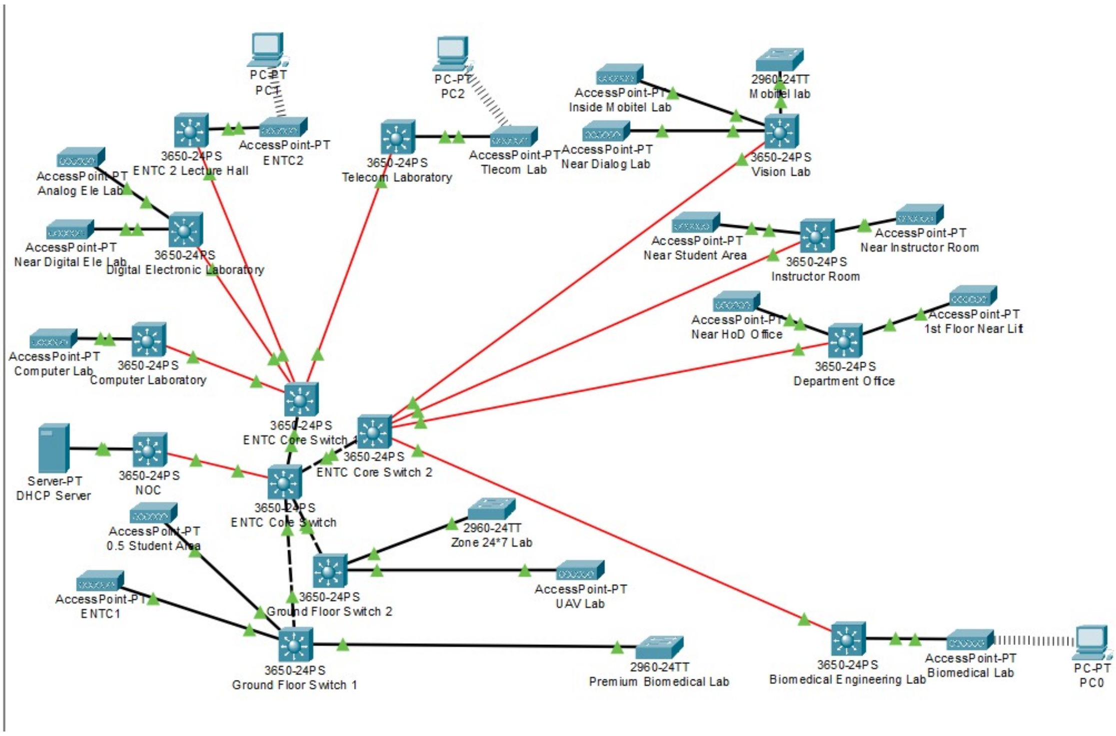

Figure 6: ENTC Internal LAN Topology

4 Link for Packet Tracer Simulation files

https://drive.google.com/drive/folders/1J-HlFWY9Mr9DaYCqmT2mKXr7IYLdDddC?usp=drive_link

5 Conclusion

A hierarchical backbone network was designed and simulated for the University of Moratuwa, connecting 13 major buildings through a partial ring topology centred on the NOC at CITeS. Single-mode fibre was selected for all inter-building links to ensure longevity and upgrade flexibility over the 20–25 year design horizon. OSPF dynamic routing provides automatic path selection and fault tolerance. The ENTC internal network was designed as a reference LAN with a three-tier hierarchy, structured cabling, and wireless coverage throughout. All connectivity was verified through Cisco Packet Tracer simulation.